

Conditional Drifting Models for Learned Channel Simulation

Rick Fritschek and Rafael F. Schaefer Chair of Information Theory and Machine Learning
Technische Universität Dresden, Germany

Abstract—Accurate stochastic channel simulation is central to data-driven communication system design, especially when channel measurements are costly to acquire repeatedly or when differentiable learned surrogates are needed inside end-to-end optimization loops. Recent work has shown that conditional diffusion models are strong channel generators, but their inference cost scales with the number of reverse denoising steps. This paper studies conditional drifting models as a low-latency alternative and evaluates them against conditional denoising diffusion probabilistic modeling (DDPM), conditional denoising diffusion implicit modeling (DDIM), and conditional Wasserstein generative adversarial network (WGAN) baselines. We evaluate residual drifting, which models $e = y - x$, direct-output drifting, which models $p(y | x)$, an enhanced direct-output variant with a conditioning-aware kernel, and a new condition-wise Sinkhorn drifting variant motivated by Wasserstein-gradient-flow one-step generation. The generator-level benchmark covers additive white Gaussian noise (AWGN), Rayleigh fading, and solid-state power amplifier (SSPA) nonlinearity; downstream channel-implant experiments additionally include a compact tapped-delay-line (TDL) wireless channel. The key modeling constraint is that a wireless channel generator does not transport the condition x : the true and generated joint laws share the same input marginal, and only the conditional output law should move. We therefore define a conditional Sinkhorn divergence by integrating Sinkhorn divergences over fixed-input conditional laws $p(\cdot | x)$, yielding a drift field that applies the Wasserstein-flow velocity separately for each transmitted-symbol condition. The reference diffusion/WGAN comparison shows that diffusion achieves the strongest sliced Wasserstein distance (SWD) overall, while direct-output drifting is the strongest one-shot generator under direct output-space evaluation and conditioning-aware kernels can improve downstream AWGN SER. The drift-field ablation shows that condition-wise Sinkhorn drifting is the strongest drifting-family symbolic-coding surrogate on AWGN, Rayleigh, SSPA, and TDL; diffusion remains the strongest learned downstream reference on the harder Rayleigh, SSPA, and TDL curve tests. On SSPA, the Sinkhorn result requires an update-budget-controlled W-Flow run because the sharp same-condition Sinkhorn field converges much earlier than the diffusion-style training preset. The results also show that global output-space SWD alone is not sufficient for selecting a channel surrogate.

Index Terms—learned channel models, conditional generative modeling, diffusion models, drifting models, optimal transport, Sinkhorn divergence, Wasserstein gradient flows, inference latency

I. INTRODUCTION

Accurate channel models are a basic requirement for modern communication system design. They are particularly important in neural and end-to-end learned systems, where the channel appears repeatedly inside training loops, decoder adaptation procedures, and differentiable autoencoder

pipelines. When a clean analytical channel model is unavailable, too restrictive, or poorly matched to measured data, generative channel surrogates become attractive because they can learn the full conditional distribution directly from samples.

This problem is not only about distributional accuracy. In many learning pipelines, the channel model is evaluated millions of times. As a result, sample quality, conditioning accuracy, computational cost, and differentiability all matter simultaneously. The question is therefore not merely whether a generative model can match the channel distribution, but whether it can do so at a cost compatible with large-scale training and deployment workflows.

Recent work on diffusion-based channel modeling has shown that conditional diffusion models can generate high-quality channel samples across standard benchmarks such as additive white Gaussian noise (AWGN), Rayleigh fading, and nonlinear amplifier channels [1]. Follow-up work has further studied robustness and speed-quality tradeoffs for diffusion-based channel generation [2], extended diffusion-based channel synthesis to high-dimensional user-specific channels [3], and proposed diffusion-driven digital-twin-style generation of statistical channel state information from sensing and location information [4]. More broadly, diffusion methods have also begun to influence adjacent wireless tasks such as high-dimensional channel estimation [5], low-complexity multiple-input multiple-output (MIMO) channel estimation [6], [7], joint channel estimation and data detection [8], and fast time-varying MIMO orthogonal frequency-division multiplexing (MIMO-OFDM) estimation [9]. This line of work is compelling because diffusion models are flexible, robust, and often more stable than earlier adversarial approaches. However, this improvement comes with a practical system cost: inference requires an iterative reverse denoising procedure, and the latency scales with the number of diffusion steps. This tradeoff is manageable when the channel generator is called occasionally, but it becomes more problematic when it is embedded in inner loops of iterative training or repeated Monte Carlo evaluations. This latency bottleneck has motivated several acceleration strategies for diffusion, including skipped or reduced-step sampling as in denoising diffusion implicit modeling (DDIM), as well as one-step or few-step alternatives such as consistency models and distillation-based approaches that compress multi-step samplers into substantially shorter generation paths [10]–[12]. This motivates the study of one-shot conditional generators for learned channel simulation. Among recently proposed generative paradigms, drifting models [13] are particularly interesting because they aim to learn the transport from a

simple latent source to the target data distribution directly in the network parameters, rather than unfold that transport as a multi-step sampling trajectory at inference time. Intuitively, diffusion spends computation at test time by progressively transforming noise into data through a reverse-time chain. Drifting, in contrast, spends computation during training: generated samples are repeatedly nudged toward the data manifold by a drift field, and the neural generator is updated so that future samples move closer to the desired distribution in a single forward pass. Once trained, sampling is therefore naturally one-shot. This perspective is also consistent with recent geometric analyses of generative dynamics that relate diffusion-like behavior to autonomous vector-field structure and marginalized noise geometry [14]. Recent work on one-step generation via Wasserstein gradient flows (W-Flow) [15] sharpens this viewpoint by replacing heuristic drifting fields with Sinkhorn-divergence optimal-transport velocities. This is directly relevant to channel simulation, because the quality and stability of the drift field is the main design question once inference is constrained to one generator evaluation.

That distinction makes the concept of drifting appealing for channel modeling. In a communication setting, the generator must be conditional on the transmitted symbol x , and it must capture not just a marginal sample cloud, but the conditional law $p(y | x)$. Because drifting is based on attraction and repulsion between generated and true samples rather than an adversarial discriminator game, it is also an interesting candidate for studying whether one-shot conditional generators can avoid some of the mode-collapse behavior often associated with adversarial training. Because drifting was introduced only very recently, no widely adopted communication-channel baseline has yet emerged. This motivates a benchmark-oriented study that reports not only comparative results, but also the concrete conditional adaptation needed to integrate drifting into a channel-learning pipeline.

The emphasis of this study is on generator-level distributional accuracy, computational tradeoffs, and downstream channel-implant checks rather than on a full communication-system design study over all modulation and coding regimes. The main contributions are summarized as follows.

- We formulate conditional drifting generators for channel simulation in both residual form, using $e = y - x$, and direct-output form, modeling $p(y | x)$ directly.
- We identify the kernel attraction–repulsion field used in the drifting baseline as an empirical approximation, include a conditioning-aware direct-output kernel variant, and derive a condition-wise Sinkhorn alternative from a conditional Wasserstein-gradient-flow objective.
- We show why a naive joint Sinkhorn drift on (x, y) is not the correct conditional flow for channel simulation: the input marginal is fixed, so transport must be constrained to the output law over each transmitted symbol.
- We compare the resulting drifting variants against conditional DDPM, conditional DDIM, and conditional WGAN reference models on AWGN, Rayleigh fading, and solid-state power amplifier (SSPA) channels, and we add a TDL wireless channel in the downstream channel-implant study.

- We quantify the accuracy–latency tradeoff and show that one-shot drifting provides large inference-time savings while remaining competitive in distributional accuracy; a symbolic AWGN case study illustrates that conditioning-aware kernel design can improve downstream direct-output drifting performance.
- We report a multi-seed drift-field ablation with direct SWD, anchor-conditioned metrics, and downstream SER/BER, and compare the distributional results against DDPM, DDIM, and WGAN wherever the protocols overlap.

Source code will be made available through the first author’s GitHub profile upon publication¹.

II. PROBLEM FORMULATION AND DRIFTING BACKGROUND

A. What drifting changes conceptually

Because drifting is very recent and remains less familiar than diffusion, we briefly summarize its generative viewpoint. Following [13], let f_θ map a latent variable $z \sim p_z$ to a sample $u = f_\theta(z)$, and let

$$q_\theta = (f_\theta)_\# p_z \quad (1)$$

denote the pushforward distribution induced by the generator. The training objective is no longer expressed as a likelihood surrogate or as a denoising score-matching loss. Instead, the paper introduces a *drifting field* that tells us how samples drawn from the current model distribution should move so that the model distribution approaches the target distribution p .

At a high level, the desired property is an equilibrium condition: if the model already matches the target distribution, then the drift should vanish. Using $V_{p,q}(u)$ to denote the field evaluated at a sample u under target distribution p and model distribution q , the fixed-point intuition can be summarized as

$$q = p \implies V_{p,q}(u) = 0, \quad (2)$$

which is the sense in which matched distributions form an equilibrium of the training dynamics [13].

This makes the training logic very different from diffusion. In diffusion, one learns a denoiser and then performs iterative refinement at inference time. In drifting, one instead constructs improved targets during training by moving current samples along the field $V_{p,q}$, and then regresses the generator onto those improved targets. As a result, the iterative correction happens during optimization rather than at test time.

The one-step view can be written as follows. Starting from a generated sample $u = f_\theta(z)$, define a drifted target

$$\tilde{u} = \text{stopgrad}(u + \eta V_{p,q_\theta}(u)), \quad (3)$$

and optimize the generator by

$$\mathcal{L}_{\text{drift}} = \mathbb{E}_{z \sim p_z} [\|f_\theta(z) - \tilde{u}\|_2^2]. \quad (4)$$

The stop-gradient is essential here: it makes the drifted sample behave like a fixed target for the current optimization step.

¹<https://github.com/Fritschek>.

Repeated training steps therefore evolve the pushforward q_θ toward the target distribution. This is also the cleanest way to see why drifting is naturally one-shot at inference: once the generator has learned to emit samples near equilibrium, sampling only requires drawing z and evaluating $f_\theta(z)$, with no reverse-time chain.

The W-Flow formulation [15] connects this one-shot training principle to entropic optimal transport. Let $\text{OT}_\varepsilon(q, p)$ denote the entropic optimal-transport cost between a generated law q and target law p , computed with Sinkhorn iterations [16]. The debiased Sinkhorn divergence is

$$S_\varepsilon(q, p) = \text{OT}_\varepsilon(q, p) - \frac{1}{2} \text{OT}_\varepsilon(q, q) - \frac{1}{2} \text{OT}_\varepsilon(p, p), \quad (5)$$

which removes the entropic self-bias and interpolates between optimal-transport and kernel-like behavior as ε changes [17]. For an empirical generated particle u , the corresponding W-Flow velocity can be written in barycentric-projection form as

$$V_\varepsilon(u) = T_{q,p}^\varepsilon(u) - T_{q,q}^\varepsilon(u), \quad (6)$$

where $T_{q,p}^\varepsilon$ maps a generated sample to the barycenter of target samples under the entropic optimal-transport coupling, and $T_{q,q}^\varepsilon$ is the generated self-transport projection. This has the same attraction–repulsion structure as the practical drifting rule, but the weights are obtained from a globally mass-constrained coupling rather than from independently row-normalized kernels.

Fig. 1 gives an actual toy particle simulation in the spirit of the particle-based demonstrations used in [13]. For two fixed channel inputs, we initialize mismatched conditional residual particle clouds and then evolve them under the same attraction–repulsion drift rule used in our channel benchmark. The figure therefore illustrates real drift dynamics rather than a hand-drawn schematic, while still serving only as a toy analogue of the full learned generator. To show that the same idea can train an explicit parameterized generator, Fig. 2 uses a tiny conditional MLP trained with the drifting objective on the same toy target.

B. Conditional channel simulation

We now specialize the generic drifting view to learned conditional channel simulation. Let $x \in \mathbb{R}^n$ denote the transmitted channel input and $y \in \mathbb{R}^n$ the channel output. The objective is to learn a conditional generator that reproduces the channel law given x . In this setting, conditioning enters through the transmitted symbol, while stochasticity is carried by a latent variable $z \sim \mathcal{N}(0, I)$.

Two target parameterizations are useful in the experiments. The first is the direct channel-output formulation, in which the generator approximates $p(y | x)$ and predicts $\hat{y} = g_\theta(x, z)$. The second is the residual formulation, in which

$$e = y - x \quad (7)$$

is modeled instead, so that the generator approximates $p(e | x)$ and the channel output is reconstructed as

$$\hat{y} = x + \hat{e}. \quad (8)$$

TABLE I
TWO TARGET PARAMETERIZATIONS FOR CONDITIONAL DRIFTING.

Mode	Output of g_θ	Target object	Short interpretation
Direct	$\hat{y}$	Learn $p(y x)$	Learns the channel law directly
Residual	$\hat{e}$, then $\hat{y} = x + \hat{e}$	Learn $p(e x)$ with $e = y - x$	Learns the channel perturbation

The direct formulation corresponds to learning the channel law itself, whereas the residual formulation injects the inductive bias that the channel output can be understood as the transmitted symbol plus a structured perturbation. In the experiments below, the direct formulation aligns naturally with direct-output diffusion and WGAN baselines, while the residual formulation is often easier to optimize but can fall short on nonlinear channels.

C. Conditional drifting model

We use a practical conditional adaptation inspired by [13]. To avoid confusion with the communication input symbol x , we use u above for the generic sample variable in the abstract formulation and reserve x below for the transmitted symbol. In our implementation, drifting can be applied either in direct-output space or in residual space. In direct mode, the generator predicts $\hat{y} = g_\theta(x, z)$; in residual mode, it predicts $\hat{e} = g_\theta(x, z)$ with reconstruction $\hat{y} = x + \hat{e}$, where

$$z \sim \mathcal{N}(0, I). \quad (9)$$

In both cases, g_θ is implemented as a two-hidden-layer multi-layer perceptron with hidden width 128, latent dimension 16, and SiLU activations:

$$[x, z] \rightarrow \text{Linear} \rightarrow \text{SiLU} \rightarrow \text{Linear} \rightarrow \text{SiLU} \rightarrow \text{Linear}. \quad (10)$$

The training target is formed from an attraction–repulsion drift field in the chosen target space. This is a concrete channel-specific instantiation of the generic field $V_{p,q}$. Given generated samples $G = \{g_i\}$ and positive samples $P = \{p_j\}$, we compute

$$V(g_i) = \left(\sum_j w_{ij}^{(P)} p_j - g_i \right) - \lambda \left(\sum_k w_{ik}^{(G)} g_k - g_i \right), \quad (11)$$

where $w^{(P)}$ and $w^{(G)}$ are row-normalized RBF kernel weights. The first term attracts each generated sample toward a local barycenter of true samples in the chosen target space, and the second term repels it from nearby generated samples to reduce collapse. This yields a detached target

$$\tilde{g}_i = \text{stopgrad}(g_i + \eta V(g_i)), \quad (12)$$

and the generator is optimized by

$$\mathcal{L}_{\text{drift}} = \frac{1}{|G|} \sum_i \|g_i - \tilde{g}_i\|_2^2. \quad (13)$$

In other words, the generic update $u \mapsto u + \eta V_{p,q}(u)$ from the drifting formulation is realized here with a kernel-based drift field estimated directly from minibatches of true and

generated samples in the selected target space. At a high level, this empirical drift field can be interpreted as a kernel particle interaction rule. The attraction term pulls generated samples toward the target sample cloud, while the repulsion term prevents generated particles from collapsing onto a few modes. In this sense, the method is related in spirit to kernel-based distribution shaping and MMD-like matching, although we use it here as a practical training rule rather than derive a formal discrepancy minimization objective.

In the simulations, the drift scale is $\eta = 1.0$, the repulsive weight is $\lambda = 1.0$, the minimum kernel bandwidth is 0.2, and the maximum drift norm is 2.0. These settings were chosen because they produced stable training behavior and materially improved the early AWGN drifting baselines.

a) *Conditioning-aware kernel variant.*: For the enhanced direct-output drifting model, we replace the target-only kernel by a kernel on joint condition-target pairs. Instead of computing sample interactions from $k(y_i, y_j)$, we compute them from $k((x_i, y_i), (x_j, y_j))$, so that attraction and repulsion depend on similarity in both the channel input and the output space. In the direct-output setting, the target kernel can also incorporate residual similarity, yielding a drift field that is more closely aligned with the conditional channel structure. The generator architecture is unchanged.

This kernel rule is the conditioning-aware baseline used for the direct-output drifting variants. It is computationally simple, but it also leaves the most important design choice—the minibatch coupling used to build V —as a channel-specific heuristic. We therefore also study Sinkhorn-based drift fields that replace the row-normalized kernel interaction with the velocity in (6), while preserving the same detached-target training loop and the same one-shot inference path.

D. Condition-wise Sinkhorn drifting

The W-Flow objective in [15] is unconditional: both p and q_θ are distributions on the generated sample space. For channel simulation, the generated and true joint laws share the same transmitted-symbol marginal. Writing μ for the input distribution,

$$p(dx, dy) = \mu(dx)p_x(dy), \quad q_\theta(dx, dy) = \mu(dx)q_{\theta,x}(dy). \quad (14)$$

The condition x is side information and should not be transported. Transporting in the full joint space (x, y) therefore solves a different problem from matching the channel law $p(y | x)$, because it allows the geometry of the fixed condition variable to influence the coupling while the update can only move the output component.

We instead define the conditional Sinkhorn functional

$$\mathcal{S}_\varepsilon^{\text{cond}}(q_\theta, p) = \int S_\varepsilon(q_{\theta,x}, p_x) \mu(dx), \quad (15)$$

where the Sinkhorn divergence is computed separately for each fixed transmitted symbol, either on y in direct-output mode or on $e = y - x$ in residual mode. In measure-theoretic terms, the conditional law $p_x = p(\cdot | x)$ is the output fiber over condition x ; throughout the paper we refer to the resulting estimator

as condition-wise Sinkhorn drifting. Equivalently, admissible couplings must preserve the condition:

$$\pi(dx, dy, d\bar{y}) = \mu(dx)\pi_x(dy, d\bar{y}), \quad \pi_x \in \Pi(q_{\theta,x}, p_x). \quad (16)$$

The Wasserstein gradient flow of (15) has no x -velocity and evolves only inside each fixed-input conditional law:

$$\partial_t q_t(x, y) + \nabla_y \cdot (q_t(x, y)v_t(x, y)) = 0. \quad (17)$$

Applying the W-Flow velocity condition-wise gives

$$v_t(x, y) = T_{q_{t,x}, p_x}^\varepsilon(y) - T_{q_{t,x}, q_{t,x}}^\varepsilon(y). \quad (18)$$

Thus, if the unconditional Sinkhorn velocity has no stationary point except equality for each fixed condition, then every stationary point of the conditional flow satisfies

$$q_{\theta,x} = p_x \quad \text{for } \mu\text{-a.e. } x. \quad (19)$$

This is the conditional analogue needed for channel simulation: the equilibrium is not merely equality of a joint sample cloud under an arbitrary joint cost, but equality of the complete family of conditional channel laws.

The empirical estimator follows the same condition-wise structure. For anchors $x_i \sim \mu$, we draw multiple generated samples $\hat{y}_{i,k} = g_\theta(x_i, z_{i,k})$, multiple positive channel samples $y_{i,j} \sim p(\cdot | x_i)$, and an independent generated reference batch $\hat{y}'_{i,\ell} = g_\theta(x_i, z'_{i,\ell})$. Sinkhorn barycentric projections are then computed separately for each anchor:

$$v_{i,k} = T_{\hat{Q}_i, P_i}^\varepsilon(\hat{y}_{i,k}) - T_{\hat{Q}_i, \hat{Q}'_i}^\varepsilon(\hat{y}_{i,k}), \quad (20)$$

followed by the detached regression target

$$\tilde{y}_{i,k} = \text{stopgrad}(\hat{y}_{i,k} + \eta v_{i,k}). \quad (21)$$

This estimator is exact for simulator channels where repeated draws at the same x_i are available. For measured channels with only one observation per condition, the same idea requires a local conditional approximation, for example by grouping nearby x -values or using condition-space kernel weights before solving the output-space Sinkhorn problem.

The major operational advantage is straightforward: once training is complete, sampling requires only a single evaluation of $g_\theta(x, z)$. In residual mode, this produces a channel residual sample $\hat{e}$ followed by reconstruction $\hat{y} = x + \hat{e}$. In direct mode, it produces $\hat{y}$ directly. There is no reverse chain, no ODE solver, and no per-sample iterative schedule at inference.

III. EXPERIMENTAL SETUP

A. Channel models

We use four stochastic channel families. All channels are implemented as real-valued maps $x \mapsto y$, but the coordinate interpretation differs by channel. AWGN and Rayleigh use real-valued vectors directly and have dimension $n = 7$ in the generator-level benchmark. SSPA and TDL use paired in-phase/quadrature coordinates with $n = 8$, corresponding to four complex symbols. Noise variables are sampled independently across real coordinates unless the deterministic channel transformation or tapped-delay convolution introduces coupling. The generator-level comparison uses AWGN, Rayleigh,

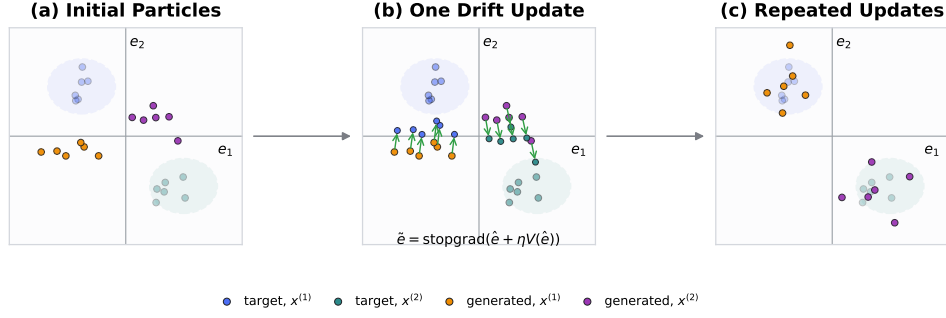


Fig. 1. Toy conditional particle simulation in residual space. Two different channel inputs induce two different target residual distributions. Starting from mismatched generated particle clouds, the middle panel applies one attraction–repulsion drift update of the form $\tilde{e} = \text{stopgrad}(\hat{e} + \eta V(\hat{e}))$, and the last panel shows the particle clouds after repeated updates. This figure is an actual toy simulation of the drift field, not a hand-drawn schematic.

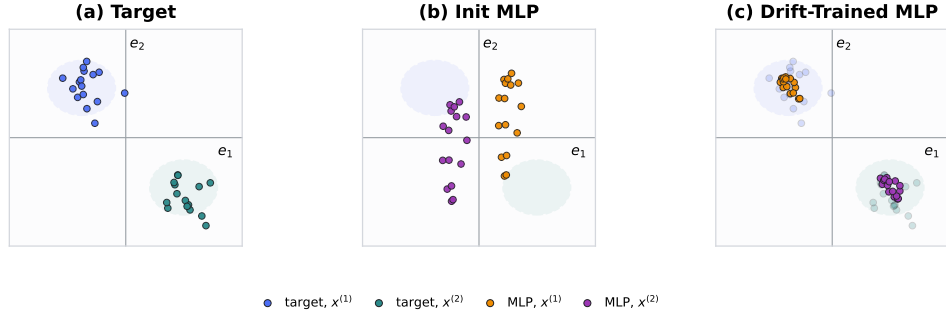


Fig. 2. Toy conditional MLP trained with the drifting objective. The left panel shows target conditional residual samples. The middle panel shows samples from a randomly initialized tiny conditional MLP $g_\theta(x, z)$. The right panel shows the same MLP after training with the drift objective, where the generated samples move close to the correct condition-dependent residual clouds.

and SSPA because these channels have matched DDPM, DDIM, WGAN, and drifting reference runs. The downstream channel-implant study additionally includes TDL as a wireless channel with memory.

AWGN:

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2 I). \quad (22)$$

This is the standard additive white Gaussian noise reference channel and serves as the simplest near-identity baseline.

Rayleigh: the transmitted symbol is multiplied elementwise by a random fading amplitude and then corrupted by additive Gaussian noise:

$$y = h \odot x + n, \quad (23)$$

where $\odot$ denotes elementwise multiplication,

$$h_k = \frac{1}{\sqrt{2}} \sqrt{u_k^2 + v_k^2}, \quad u_k, v_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad (24)$$

and $n \sim \mathcal{N}(0, \sigma^2 I)$.

Each real component is therefore scaled by an independent Rayleigh-distributed magnitude before noise is added. This channel is multiplicative rather than additive and is correspondingly less aligned with a pure residual perturbation view. As implemented here, Rayleigh is also a real-valued vector channel rather than a paired-I/Q complex channel.

SSPA: the input is passed through a smooth solid-state power amplifier nonlinearity before additive noise is applied.

In this case, the real-valued representation is interpreted as paired I/Q components. Let

$$r = \sqrt{x_I^2 + x_Q^2} \quad (25)$$

denote the input amplitude. The nonlinear amplitude law used in the benchmark is

$$a(r) = \frac{G}{\left(1 + \left(\frac{Gr}{A_0}\right)^{2p}\right)^{1/(2p)}}, \quad (26)$$

with default parameters $p = 3$, $A_0 = 1.5$, and $G = 5.0$. The noiseless amplifier output is $\tilde{x} = a(r)x$, and the observed channel output is

$$y = \tilde{x} + n, \quad n \sim \mathcal{N}\left(0, \frac{\sigma^2}{2} I\right). \quad (27)$$

This channel introduces amplitude-dependent nonlinear distortion while preserving the input direction in the I/Q plane.

TDL: The TDL channel is a compact short-block approximation of the 3GPP TR 38.901 TDL-D profile [18]. The real vector is interpreted as a complex codeword $s_m = x_{2m} + jx_{2m+1}$, $m = 0, \dots, N_c - 1$, with $N_c = 4$ complex channel uses in the symbolic coding runs. For each codeword, the channel samples a finite set of complex taps and applies circular convolution,

$$\tilde{s}_m = \sum_{\ell=0}^{L_h-1} h_\ell s_{(m-d_\ell) \bmod N_c}. \quad (28)$$

The observed output is

$$y_m = \tilde{s}_m + n_m, \quad n_m \sim \mathcal{CN}(0, \sigma^2). \quad (29)$$

The implemented profile uses the TDL-D line-of-sight structure with first-tap K -factor 13.3 dB. Because the codeword is short, the normalized path delays from the standard profile are rounded to a small set of effective circular shifts and path powers are renormalized after aggregation. This keeps the channel compatible with the short symbolic autoencoder while preserving the essential feature that the output is a random multipath mixture rather than a memoryless componentwise perturbation.

B. Training protocol

The experiments are organized into two complementary protocols. First, the generator-level protocol compares the proposed drifting family against the established conditional DDPM, DDIM, and WGAN references on the three channels for which matched baseline runs are reported here: AWGN, Rayleigh, and SSPA. Second, the drift-field ablation holds the drifting network fixed and compares four training fields: target-space kernel drifting, conditioning-aware joint-kernel drifting, joint Sinkhorn drifting, and condition-wise Sinkhorn drifting. This second protocol is reported over 100 seeds for the main drift-field runs and is evaluated with direct SWD, anchor-conditioned metrics, and downstream symbolic SER/BER. For SSPA, we additionally report a 30-seed update-budget-controlled condition-wise Sinkhorn run because the corrected same-condition transport field saturates after far fewer optimizer updates than the long diffusion-style SSPA preset. The TDL channel is included in the drift-field study and in the downstream SER curves with diffusion and WGAN references; it is omitted only from the direct-output generator-level SWD reference table.

a) Channel settings.: For AWGN and Rayleigh, the setting is $n = 7$, while for SSPA and TDL it is $n = 8$. The downstream symbolic autoencoder uses message-alphabet size $M_{\text{msg}} = 16$ for AWGN, Rayleigh, and TDL, and $M_{\text{msg}} = 64$ for SSPA. Thus $R = \log_2(M_{\text{msg}})/n$, giving rates $4/7$, $4/7$, $6/8$, and $4/8$, respectively. The corresponding $(E_b/N_0, R)$ pairs are (5 dB, $4/7$) for AWGN, (12 dB, $4/7$) for Rayleigh, (8 dB, $6/8$) for SSPA, and (10 dB, $4/8$) for TDL. These settings yield noise standard deviations 0.5260, 0.2350, 0.3251, and 0.3162, respectively.

b) Data and optimization settings.: For the generator-level AWGN and Rayleigh runs, we use training dataset size 10^7 , batch size 5000, and 30 epochs for drifting, diffusion, and WGAN. For SSPA, we use training dataset size 10^7 , batch size 4096, and 160 epochs for the diffusion, WGAN, and reference drifting-family runs. The reported SSPA condition-wise Sinkhorn row uses 120,000 samples, batch size 4096, and 160 epochs, corresponding to approximately 4.8k optimizer updates; this compact budget was selected because the corrected condition-wise W-Flow field had already converged, whereas continuing to the full 10^7 -sample budget over-optimized the sharp same-condition field. For these three channels, generator metrics are evaluated on 10^6 samples

with 128 SWD projections. The TDL drift-field ablation uses the same compact budget as the nonlinear short-block ablations: dataset size 120,000, batch size 512, 60 drifting epochs, evaluation size 100,000, and 128 SWD projections. Downstream symbolic autoencoders are trained through the analytic channel or through the learned channel implant and are always evaluated on the analytic channel.

c) Model settings.: Drifting uses a conditional MLP with latent dimension 16, hidden width 128, two hidden layers, SiLU activations, and Adam with learning rate 10^{-3} . The base drifting objective uses drift scale 1.0, median-heuristic bandwidth selection, minimum bandwidth 0.2, maximum drift norm 2.0, and repulsive weight 1.0. We train both parameterizations: direct drifting with target y and residual drifting with target $e = y - x$. The four drift-field variants use the same direct-output generator architecture and detached-target update, but replace or modify the training drift field. The target-kernel row uses only target-space similarities, the joint-kernel row uses the conditioning-aware kernel, the joint Sinkhorn row uses a Sinkhorn/W-Flow drift in joint condition-output features, and the condition-wise row uses the per-anchor conditional Sinkhorn field in (18). Unless otherwise stated, Sinkhorn variants use 10 Sinkhorn iterations, automatic entropic scale selection with minimum $\varepsilon = 10^{-3}$, four generated samples, four positive channel samples, and four generated reference samples per condition anchor.

For AWGN, Rayleigh, and SSPA, diffusion models $p(y | x)$ directly with v -prediction, cosine-ZF variance schedule, 100 diffusion steps, and EMA decay 0.9. Hidden widths are 110 (AWGN), 128 (Rayleigh), and 110 (SSPA). AWGN and Rayleigh use a two-stage learning-rate schedule (10 epochs at 10^{-3} , then 20 epochs at 10^{-4}), while SSPA uses a constant 10^{-4} for 160 epochs. We report DDPM and DDIM with 100, 50, 20, and 10 denoising steps.

WGAN uses a conditional Wasserstein objective with RM-Sprop at 10^{-4} for generator and critic, five critic updates per generator update, and weight clipping at 0.01. Hidden widths are 128 (AWGN) and 256 (Rayleigh/SSPA). WGAN is reported with residual-space SWD in the residual benchmark and direct-space SWD where the direct-output protocol is used.

d) Experimental fairness.: The benchmark is intended as a practical comparative study rather than a strict matched-capacity comparison. Architectures and training schedules were chosen as stable implementations for each model family, rather than enforcing identical parameter counts or compute budgets across drifting, diffusion, and WGAN. The Sinkhorn rows are therefore reported as a drift-field ablation under a matched drifting architecture and training budget. They are compared to diffusion and WGAN wherever the observable metric and channel protocol overlap, but the downstream SER/BER table is limited to the learned drifting/Sinkhorn surrogates and the analytic-channel reference floor.

As an additional diagnostic, Table II reports inference-network parameter counts used in this benchmark.

e) Seeding and reporting.: The diffusion/WGAN reference benchmark results are reported as mean $\pm$ standard deviation over ten seeds. The Sinkhorn drift-field ablation is

TABLE II
MODEL PARAMETER COUNTS IN THE BENCHMARK SETUP. COUNTS REFER TO INFERENCE-TIME NETWORKS. DDPM AND DDIM SHARE THE SAME DENOISER WEIGHTS. WGAN ROWS REPORT GENERATOR-SIDE PARAMETERS.

Model	AWGN	Rayleigh / SSPA
Drifting (direct / residual)	20,487	20,487 / 20,744
Diffusion denoiser (DDPM / DDIM)	59,847	74,247 / 60,178
WGAN generator	19,335	71,431 / 72,200

reported over 100 seeds unless a table note states otherwise; the SSPA update-budget check uses 30 seeds. W-Flow tables use mean $\pm$ standard error to show the uncertainty of the seed average. Each run uses deterministic initialization to reduce run-to-run variance unrelated to model stochasticity.

C. Evaluation metric

The diffusion/WGAN/drifting reference comparison emphasizes generator-level distributional accuracy and timing, while the Sinkhorn drift-field ablation adds anchor-conditioned diagnostics and downstream symbolic SER/BER. The primary metric is sliced Wasserstein distance (SWD), computed in the target space used by each benchmark path. For direct-output models, we report $\text{SWD}(\{\hat{y}_i\}, \{y_i\})$. For residual models, we report $\text{SWD}(\{\hat{e}_i\}, \{e_i\})$. Given true samples $P = \{v_i\}_{i=1}^N$ and generated samples $Q = \{\hat{v}_i\}_{i=1}^N$, SWD is estimated by drawing random unit directions θ_m , projecting both clouds onto those directions, computing the one-dimensional Wasserstein distance for each projection, and averaging over projections:

$$\text{SWD}(P, Q) = \frac{1}{M} \sum_{m=1}^M W_1(\{\langle \theta_m, v_i \rangle\}_{i=1}^N, \{\langle \theta_m, \hat{v}_i \rangle\}_{i=1}^N).$$

Thus SWD compares the generated and true sample clouds through many one-dimensional views and captures geometric mismatch more faithfully than a purely bin-based marginal comparison. In our experiments, SWD is estimated using 128 random one-dimensional projections on AWGN, Rayleigh, and SSPA. SWD is used primarily for comparability with prior diffusion-channel benchmarks [1], [2]. Compared with histogram-based L_1 distance, it also better captures sample geometry in $\mathbb{R}^n$. While SWD provides a geometry-aware comparison of conditional output distributions, it does not directly capture decision-relevant behavior such as error rates, which depend on the joint structure of (x, y) . For communication use, two generators with similar global sample-cloud discrepancy can induce different conditional means, conditional covariance errors, and downstream symbol-error rates. The Sinkhorn drift-field study therefore also includes anchor-conditioned moment metrics and symbolic channel-implant SER/BER checks when evaluating the proposed drifting variants.

IV. BENCHMARK RESULTS

A. Unified SWD benchmark across channels

Table III reports a unified direct-output SWD comparison across AWGN, Rayleigh, and SSPA. The reference rows

summarize the diffusion and WGAN baselines; the drifting-family rows contain direct drifting and the proposed drift-field variants under a matched one-shot architecture. DDIM-100 gives the lowest direct-space SWD on AWGN, Rayleigh, and SSPA. Among the W-Flow variants, condition-wise Sinkhorn is strongest under direct SWD on AWGN and Rayleigh and is numerically close to the best SSPA W-Flow rows under the compact update-budget run. Within the full drifting-family block, condition-wise Sinkhorn is lowest on AWGN and Rayleigh, while direct drifting remains lowest on SSPA under this global metric. This already indicates that no single drift field is uniformly optimal across channel geometry.

Timing follows the expected pattern: drifting and WGAN operate at roughly $0.1 \mu\text{s}$ per sample on the local CUDA timing setup, whereas DDPM and DDIM require microsecond-to-tens-of-microseconds sampling times depending on the sampler and the number of denoising steps. Thus, even when diffusion achieves the strongest SWD, one-shot generators retain a substantial practical latency advantage. The W-Flow variants retain the same one-shot inference path as direct drifting; their cost differences are training-time drift-field costs. Table IV therefore reports W-Flow ablation rows under the same 2% local CUDA timing protocol. On AWGN, Rayleigh, and SSPA, the condition-wise Sinkhorn row is faster to train than the global kernel rows because it solves many small same-condition transport problems instead of forming a global minibatch interaction matrix.

B. Metric-space check for residual baselines

This diagnostic is separate from the direct-output SWD ranking above. To quantify the effect of the evaluation space, we compare residual drifting and WGAN under both direct y -space SWD and residual-space SWD in Table V. WGAN changes little between the two metrics, whereas residual drifting degrades substantially in direct y -space, especially on Rayleigh and SSPA. This indicates that residual drifting is well aligned with residual modeling, but that its advantage does not transfer directly to full output-space matching. The Sinkhorn ablation is therefore reported in direct-output, anchor-conditioned, and downstream coding metrics rather than in residual space alone.

C. Sinkhorn drift-field interpretation

We next evaluate the W-Flow-inspired variants over the completed multi-seed drift-field ablation. The compared variants are: the target-space kernel baseline, the conditioning-aware joint kernel, a naive joint Sinkhorn drift using (x, y) -features, and the proposed condition-wise Sinkhorn drift. All four are one-shot generators at inference time; the difference is only the training-time drift field.

The drifting-family block in Table III compares direct drifting and the proposed drift-field variants against the diffusion and WGAN reference rows. The comparison should be read with the reporting convention in mind: the DDPM/DDIM/WGAN and direct-drifting rows are ten-seed values, while the W-Flow rows are 100-seed estimates except for the marked SSPA update-budget check. Diffusion remains

TABLE III

DIRECT-OUTPUT SWD COMPARISON FOR DIFFUSION/WGAN REFERENCES AND DRIFTING VARIANTS. LOWER IS BETTER. REFERENCE ROWS REPORT THE TEN-SEED MEAN $\pm$ STANDARD DEVIATION FROM THE DIFFUSION/WGAN BENCHMARK. THE DRIFTING-FAMILY ROWS CONTAIN DIRECT DRIFTING FROM THE SAME TEN-SEED BENCHMARK AND THE W-FLOW DRIFT-FIELD ABLATION; W-FLOW ROWS REPORT ALL-SEED MEAN $\pm$ STANDARD ERROR. BOLD MARKS THE BEST DRIFTING-FAMILY ROW PER CHANNEL UNDER THE REPORTED MEAN.

Method	AWGN	Rayleigh	SSPA
<i>Diffusion/WGAN reference baselines</i>			
WGAN	0.0198 $\pm$ 0.0050	0.0171 $\pm$ 0.0050	0.0287 $\pm$ 0.0156
DDPM	0.0070 $\pm$ 0.0004	0.0079 $\pm$ 0.0008	0.0032 $\pm$ 0.0005
DDIM-100	0.0042 $\pm$ 0.0006	0.0044 $\pm$ 0.0007	0.0024 $\pm$ 0.0004
<i>Drifting family</i>			
Direct drifting	0.0100 $\pm$ 0.0007	0.0085 $\pm$ 0.0012	0.0060 $\pm$ 0.0008
Kernel target	0.0106 $\pm$ 0.0001	0.0099 $\pm$ 0.0001	0.0070 $\pm$ 0.0001
Kernel joint	0.0290 $\pm$ 0.0004	0.0492 $\pm$ 0.0002	0.0074 $\pm$ 0.0001
Joint Sinkhorn	0.0148 $\pm$ 0.0002	0.0152 $\pm$ 0.0002	0.0117 $\pm$ 0.0002
Condition-wise Sinkhorn	0.0058 $\pm$ 0.0001	0.0073 $\pm$ 0.0001	0.0071 $\pm$ 0.0002*

* The SSPA condition-wise Sinkhorn value uses the corrected 30-seed update-budget-controlled run (120,000 samples, batch 4096, 160 epochs), where the one-shot field saturated after roughly 4.8k optimizer updates.

TABLE IV

PROJECTED TRAINING AND INFERENCE TIMING ON AN NVIDIA RTX 5060 Ti GPU. TRAINING HOURS ARE EXTRAPOLATED FROM A 2% TIMING RUN; INFERENCE IS REPORTED AS TIME PER SAMPLE.

Method	AWGN			Rayleigh			SSPA			TDL		
	Train [h]	Inf.	Total [h]	Train [h]	Inf.	Total [h]	Train [h]	Inf.	Total [h]	Train [h]	Inf.	Total [h]
<i>One-shot generators</i>												
Drifting (res.)	1.309	0.131 us	1.310	1.300	0.123 us	1.300	5.656	0.121 us	5.656	0.013	0.124 us	0.013
Drifting (dir.)	1.299	0.114 us	1.300	1.299	0.117 us	1.300	5.656	0.118 us	5.656	0.007	0.116 us	0.007
Joint-kernel drift	1.863	0.116 us	1.864	1.864	0.116 us	1.864	8.112	0.119 us	8.112	0.008	0.115 us	0.008
Joint Sinkhorn	0.917	0.116 us	0.918	0.919	0.115 us	0.920	3.956	0.117 us	3.956	0.010	0.116 us	0.010
Condition-wise Sinkhorn	0.048	0.115 us	0.048	0.048	0.115 us	0.049	0.296	0.117 us	0.297	0.009	0.115 us	0.009
WGAN	0.064	0.108 us	0.064	0.085	0.108 us	0.086	0.478	0.107 us	0.478	0.022	0.108 us	0.022
<i>Diffusion samplers</i>												
DDPM	0.020	0.046 ms	0.147	0.020	0.045 ms	0.145	0.124	0.047 ms	0.253	0.005	0.045 ms	0.007
DDIM-100	0.020	0.039 ms	0.128	0.020	0.039 ms	0.128	0.124	0.039 ms	0.231	0.005	0.039 ms	0.006
DDIM-50	0.020	0.019 ms	0.073	0.020	0.019 ms	0.073	0.124	0.019 ms	0.177	0.005	0.019 ms	0.006
DDIM-20	0.020	0.008 ms	0.042	0.020	0.008 ms	0.042	0.124	0.008 ms	0.145	0.005	0.008 ms	0.006
DDIM-10	0.020	0.004 ms	0.031	0.020	0.004 ms	0.031	0.124	0.004 ms	0.135	0.005	0.004 ms	0.005

All rows use the same 2% extrapolation protocol. In the one-shot block, bold total times mark the lowest projected total for each channel. The W-Flow rows use the same one-shot generator architecture as direct drifting; their training-time differences come only from the drift-field computation.

TABLE V

DIRECT- AND RESIDUAL-SPACE SWD FOR RESIDUAL DRIFTING AND WGAN. VALUES ARE TEN-SEED MEAN AND STANDARD DEVIATION; LOWER IS BETTER.

Method	Direct SWD	Residual SWD
<i>AWGN</i>		
Drifting (res.)	0.0670 $\pm$ 0.0138	0.0103 $\pm$ 0.0010
WGAN	0.0198 $\pm$ 0.0050	0.0195 $\pm$ 0.0051
<i>Rayleigh</i>		
Drifting (res.)	0.3681 $\pm$ 0.0031	0.0063 $\pm$ 0.0011
WGAN	0.0171 $\pm$ 0.0050	0.0170 $\pm$ 0.0050
<i>SSPA</i>		
Drifting (res.)	0.4812 $\pm$ 0.0022	0.0128 $\pm$ 0.0007
WGAN	0.0287 $\pm$ 0.0156	0.0269 $\pm$ 0.0162

the strongest direct-SWD reference on AWGN, Rayleigh, and SSPA. Among the W-Flow variants, condition-wise Sinkhorn gives the best direct SWD on AWGN and Rayleigh. On SSPA, the global direct-SWD means of the best W-Flow rows are close, while the compact condition-wise run is substantially better than continuing the corrected field to the full diffusion-

style budget. Direct drifting remains a useful global-SWD reference inside the drifting family, but the downstream checks below show that global SWD alone is not enough to select a channel surrogate. This is already a warning that the single global sample-cloud metric is not fully aligned with conditional channel usefulness.

D. Conditional metrics and symbolic coding

Table VI reports the corresponding anchor-conditioned diagnostics. The anchor metrics draw repeated outputs at fixed transmitted symbols and therefore probe the conditional law more directly than global SWD. Direct SWD is kept in Table III, and downstream BER/SER is kept in Table VII; this avoids reporting the same quantities in multiple tables. This changes the interpretation on several channels. For AWGN, Rayleigh, SSPA, and TDL, condition-wise Sinkhorn is the strongest drifting-family surrogate under the downstream BER/SER checks reported here. The full curve comparison below shows that diffusion remains the strongest learned reference on Rayleigh, SSPA, and TDL. On SSPA, the anchor-

conditioned metrics also explain why the compact update-budget run is preferable: the same-condition Sinkhorn field matches fixed-input output clouds well, while the full-budget corrected run over-optimizes the sharp field. This indicates that condition-wise transport is useful, but its optimizer budget must respect the geometry of saturated nonlinear channels.

The downstream symbolic results in Table VII use the learned channel surrogate for autoencoder training and the analytic channel for final evaluation. The analytic row is therefore a reference floor, not a learned model. We treat SER as the primary symbolic metric because the autoencoder message alphabet is finite; BER is retained in the table as an additional bit-level diagnostic. Within the W-Flow ablation table, condition-wise Sinkhorn is the most reliable drifting variant and fixes the SSPA failure mode once the update budget is matched to the sharp same-condition field. The full SER curves in Fig. 3 add WGAN and diffusion references to this downstream comparison. At the nominal training point, condition-wise Sinkhorn nearly reaches the analytic AWGN floor, while diffusion is the strongest learned baseline on Rayleigh, SSPA, and TDL. On SSPA, the $M_{\text{msg}} = 64$ update-budget screen gives condition-wise Sinkhorn SER 8.13×10^{-5} at $E_b/N_0 = 8$ dB, compared with 7.50×10^{-5} for WGAN, 3.13×10^{-5} for diffusion, and 1.57×10^{-5} for analytic-channel training.

For SSPA, we follow the diffusion-channel setup of [1] more closely and use message alphabet $M_{\text{msg}} = 64$, $n = 8$, rate $6/8$, and training at $E_b/N_0 = 8$ dB. The 30-seed SSPA screen in Table VII shows that the corrected condition-wise Sinkhorn implementation is reliable once the W-Flow update budget is matched to the observed convergence of the sharp same-condition field. Fig. 3 reports SER over an E_b/N_0 grid for AWGN, Rayleigh, SSPA, and TDL, which also shows where diffusion retains an advantage over one-shot drifting in downstream coding performance.

E. Long-block TurboAE check

The symbolic autoencoder results above use short code-words, so we additionally run a longer-block neural-coding check with a CNN TurboAE-style encoder-decoder at block length 64, rate $1/2$, and AWGN training point $E_b/N_0 = 4$ dB. The TurboAE models are trained either through the analytic AWGN channel or through a condition-wise Sinkhorn AWGN surrogate and are then evaluated on the analytic AWGN channel. This experiment is not intended to replace the short-block multi-channel SER/BER study; it checks whether the learned one-shot channel implant remains usable in a longer differentiable coding loop.

Fig. 4 shows the resulting BER and BLER curves over 30 seeds. At the 4 dB training point, analytic-channel training gives BER 1.96×10^{-3} with 95% CI 5.21×10^{-4} and BLER 8.18×10^{-2} with 95% CI 2.30×10^{-2} . Training through the condition-wise Sinkhorn surrogate gives BER 3.30×10^{-3} with 95% CI 4.30×10^{-4} and BLER 1.39×10^{-1} with 95% CI 1.91×10^{-2} . Thus the surrogate has a measurable long-block gap, but it does not cause a collapse of TurboAE training.

V. DISCUSSION

A. Metric-space dependence

The reference benchmark already shows that direct and residual drifting behave differently across channel classes and evaluation spaces. The residual-vs-direct diagnostic makes this explicit: residual drifting can be excellent in residual space while being poor as a full output-space model of $p(y | x)$. The W-Flow run extends this message. Kernel target, joint kernel, joint Sinkhorn, and condition-wise Sinkhorn can swap order depending on whether the metric is global SWD, anchor-conditioned SWD, or downstream BER/SER. We therefore report direct-output SWD, residual SWD, anchor-conditioned moment metrics, and downstream coding metrics side by side rather than selecting hyperparameters from a single global SWD score.

B. Conditional Sinkhorn scope

The condition-wise estimator is directly applicable to analytic or simulator channels because repeated outputs can be drawn at the same transmitted symbol. Measured channels are harder: with only one observation per x , the exact conditional law $p(\cdot | x)$ is not observable at a point. For that regime, the appropriate practical variant is local conditional Sinkhorn, where nearby conditions are grouped or kernel-weighted and the Sinkhorn problem is solved only inside a local condition block. Practical channel simulators may also need transport features that encode phase, amplitude, residual structure, or decoder-relevant projections.

VI. CONCLUSION

This paper presented a comprehensive experimental study of conditional drifting models for learned channel simulation and compared them against conditional diffusion and WGAN baselines on AWGN, Rayleigh, and SSPA channels. The results show a clear tradeoff between fidelity and latency. Diffusion provides the strongest distributional accuracy overall under global SWD, while one-shot drifting remains competitive and retains substantially lower inference cost. Under direct output-space evaluation, direct-output drifting is the strongest one-shot generator among the non-diffusion reference models, whereas the residual-vs-direct diagnostic shows that residual drifting remains strong only when evaluated in its native residual target space. The W-Flow-inspired ablation shows that the drift-field design problem is genuinely conditional: transporting the joint law of (x, y) is not equivalent to transporting $p(y | x)$ with a fixed input marginal. The completed Sinkhorn study confirms that condition-wise Sinkhorn is the strongest drifting-family symbolic-coding surrogate on AWGN, Rayleigh, SSPA, and TDL under the reported channel-specific coding setups. At the same time, the full downstream curves show that diffusion still provides the strongest learned reference on Rayleigh, SSPA, and TDL. The SSPA study is useful because it separates the transport objective from the optimizer budget: the corrected same-condition Sinkhorn field works well after convergence, but the full diffusion-style SSPA preset can over-optimize this sharper one-step field.

TABLE VI

ANCHOR-CONDITIONED DIAGNOSTICS FOR W-FLOW VARIANTS. ANCHOR SWD AND GAUSSIAN-W2 ARE COMPUTED FROM REPEATED SAMPLES AT FIXED CHANNEL INPUTS. DIRECT SWD AND BER/SER ARE REPORTED SEPARATELY IN TABLES III AND VII. VALUES ARE SEED MEANS; LOWER IS BETTER.

Channel	Variant	Anchor SWD	Anchor GW2
AWGN	Kernel target	0.3711	1.2565
AWGN	Kernel joint	0.1425	0.5936
AWGN	Joint Sinkhorn	0.1199	0.4443
AWGN	Condition-wise Sinkhorn	0.1154	0.4277
Rayleigh	Kernel target	0.3757	1.2869
Rayleigh	Kernel joint	0.1994	0.8198
Rayleigh	Joint Sinkhorn	0.1723	0.6929
Rayleigh	Condition-wise Sinkhorn	0.1097	0.3820
SSPA	Kernel target	0.2118	0.7631
SSPA	Kernel joint	0.2135	0.7685
SSPA	Joint Sinkhorn	0.2174	0.7876
SSPA	Condition-wise Sinkhorn	0.0566	0.2376
TDL	Kernel target	0.3155	1.1632
TDL	Kernel joint	0.2934	1.0939
TDL	Joint Sinkhorn	0.3221	1.1871
TDL	Condition-wise Sinkhorn	0.1054	0.4444

The SSPA condition-wise Sinkhorn row uses the same 30-seed update-budget-controlled run as Table III.

TABLE VII

DOWNSTREAM SYMBOLIC CODING METRICS FOR W-FLOW CHANNEL SURROGATES. AUTOENCODERS ARE TRAINED THROUGH EACH CHANNEL SURROGATE AND EVALUATED ON THE ANALYTIC CHANNEL. VALUES ARE MEAN $\pm$ STANDARD ERROR OVER SEEDS. LOWER IS BETTER; BOLD MARKS THE BEST LEARNED SURROGATE IN EACH ROW, WHILE THE ANALYTIC CHANNEL IS THE REFERENCE FLOOR.

Channel	Metric	Analytic	Kernel target	Kernel joint	Joint Sinkhorn	Condition-wise Sinkhorn
AWGN	BER	$0.00293 \pm 4.0 \times 10^{-5}$	$0.0187 \pm 5.7 \times 10^{-4}$	$0.0124 \pm 3.8 \times 10^{-4}$	$0.00388 \pm 5.5 \times 10^{-5}$	$0.00302 \pm 4.2 \times 10^{-5}$
AWGN	SER	$0.00551 \pm 7.3 \times 10^{-5}$	0.0351 ± 0.001	$0.0233 \pm 7.1 \times 10^{-4}$	$0.00729 \pm 9.7 \times 10^{-5}$	$0.00568 \pm 7.5 \times 10^{-5}$
Rayleigh	BER	$0.00273 \pm 4.3 \times 10^{-5}$	$0.0359 \pm 7.1 \times 10^{-4}$	0.111 ± 0.0017	$0.0177 \pm 3.3 \times 10^{-4}$	$0.00471 \pm 8.2 \times 10^{-5}$
Rayleigh	SER	$0.00514 \pm 8.0 \times 10^{-5}$	0.0685 ± 0.0012	0.208 ± 0.0032	$0.0331 \pm 5.6 \times 10^{-4}$	$0.00881 \pm 1.4 \times 10^{-4}$
SSPA	BER	$6.72 \times 10^{-6} \pm 1.9 \times 10^{-6}$	0.0661 ± 0.0046	—	—	$4.90 \times 10^{-5} \pm 7.5 \times 10^{-6}$
SSPA	SER	$1.30 \times 10^{-5} \pm 3.5 \times 10^{-6}$	0.134 ± 0.0092	—	—	$9.33 \times 10^{-5} \pm 1.4 \times 10^{-5}$
TDL	BER	$0.00158 \pm 2.1 \times 10^{-5}$	$0.0102 \pm 2.6 \times 10^{-4}$	$0.0098 \pm 2.5 \times 10^{-4}$	$0.0107 \pm 3.7 \times 10^{-4}$	$0.00293 \pm 4.4 \times 10^{-5}$
TDL	SER	$0.00297 \pm 3.6 \times 10^{-5}$	$0.0194 \pm 4.9 \times 10^{-4}$	$0.0185 \pm 4.6 \times 10^{-4}$	$0.02 \pm 6.3 \times 10^{-4}$	$0.00546 \pm 7.6 \times 10^{-5}$

SSPA uses the 30-seed $M_{\text{msg}} = 64$ update-budget screen; dashes mark variants not rerun under that SSPA coding setup.

Across the full set of results, global SWD, anchor-conditioned metrics, and downstream coding metrics do not always agree. This makes metric choice a central part of learned channel simulation rather than a post-processing detail, and motivates future channel-specific transport features or local conditional Sinkhorn approximations for measured channels.

REFERENCES

- [1] M. Kim, R. Fritschek, and R. F. Schaefer, “Diffusion models for accurate channel distribution generation,” *arXiv preprint arXiv:2309.10505*, 2023.
- [2] —, “Robust generation of channel distributions with diffusion models,” in *IEEE International Conference on Communications (ICC)*, 2024.
- [3] T. Lee, J. Park, H. Kim, and J. G. Andrews, “Generating high dimensional user-specific wireless channels using diffusion models,” *arXiv preprint arXiv:2409.03924*, 2024.
- [4] X. Gong, X. Liu, A. A. Lu, X. Gao, X. G. Xia, C.-X. Wang, and X. You, “Digital twin of channel: Diffusion model for sensing-assisted statistical channel state information generation,” *IEEE Transactions on Wireless Communications*, vol. 24, no. 5, pp. 3805–3821, 2025.
- [5] X. Zhou, L. Liang, J. Zhang, P. Jiang, Y. Li, and S. Jin, “Generative diffusion models for high dimensional channel estimation,” *IEEE Transactions on Wireless Communications*, vol. 24, no. 7, pp. 5840–5854, 2025.
- [6] B. Fesl, M. Baur, F. Strasser, M. Joham, and W. Utschick, “Diffusion-based generative prior for low-complexity MIMO channel estimation,” *IEEE Wireless Communications Letters*, vol. 13, no. 12, pp. 3493–3497, 2024.
- [7] Z. Chen, H. Shin, and A. Nallanathan, “Generative diffusion model-based variational inference for MIMO channel estimation,” *IEEE Transactions on Communications*, vol. 73, no. 10, pp. 9254–9269, 2025.
- [8] N. Zilberstein, A. Swami, and S. Segarra, “Joint channel estimation and data detection in massive MIMO systems based on diffusion models,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
- [9] H. Fu, W. Si, and R. Liu, “Conditional denoising diffusion-based channel estimation for fast time-varying MIMO-OFDM systems,” *Digital Signal Processing*, vol. 164, p. 105283, 2025.
- [10] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *International Conference on Learning Representations*, 2021.
- [11] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, “Consistency models,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 32 211–32 252. [Online]. Available: <https://proceedings.mlr.press/v202/song23a.html>
- [12] T. Yin, M. Gharbi, R. Zhang, E. Shechtman, F. Durand, W. T. Freeman, and T. Park, “One-step diffusion with distribution matching distillation,” in *CVPR*, 2024.
- [13] M. Deng, H. Li, T. Li, Y. Du, and K. He, “Generative modeling via drifting,” *arXiv preprint arXiv:2602.04770*, 2026.
- [14] S. Sahraee-Ardakan, M. Delbracio, and P. Milanfar, “The geometry of noise: Why diffusion models don’t need noise conditioning,” *arXiv preprint arXiv:2602.18428*, 2026.
- [15] J. Han, P. Li, Q. Guo, R. Xu, S. Ermon, and E. J. Candès, “One-step generative modeling via wasserstein gradient flows,” *arXiv preprint arXiv:2605.11755*, 2026.
- [16] M. Cuturi, “Sinkhorn distances: Lightspeed computation of optimal

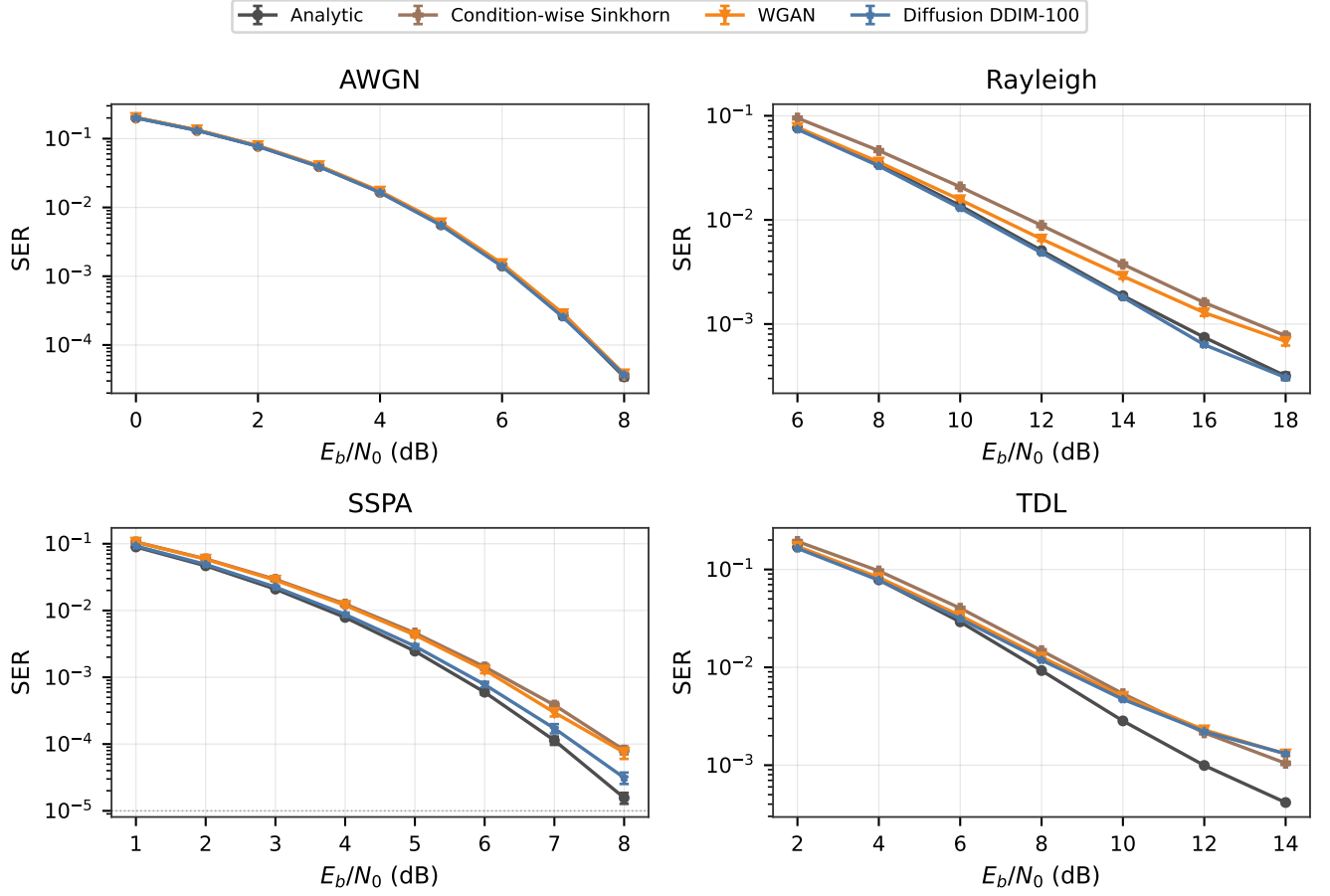


Fig. 3. **SER curves for learned channel surrogates.** Symbolic autoencoders are trained through each surrogate at the nominal training point and evaluated on the analytic channel over an E_b/N_0 grid for AWGN, Rayleigh, SSPA, TDL. Curves report seed means with standard-error bars; points below the single-run Monte Carlo resolution are clipped to the dotted reporting floor and marked as upper bounds.

transport,” in *Advances in Neural Information Processing Systems*, vol. 26, 2013, pp. 2292–2300.

- [17] J. Feydy, T. Séjourné, F.-X. Vialard, S.-i. Amari, A. Trounev, and G. Peyré, “Interpolating between optimal transport and MMD using sinkhorn divergences,” in *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 89. PMLR, 2019, pp. 2681–2690. [Online]. Available: <https://proceedings.mlr.press/v89/feydy19a.html>
- [18] 3GPP, “Study on Channel Model for Frequencies from 0.5 to 100 GHz,” 3GPP, Technical Report TR 38.901, 2022, version 17.1.0.

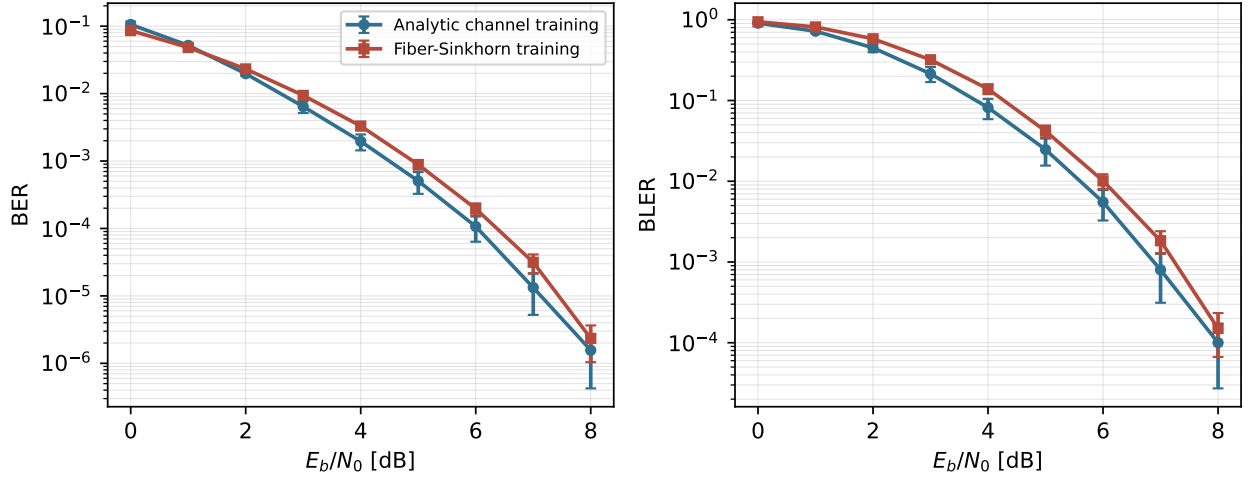


Fig. 4. **Long-block TurboAE AWGN check.** CNN TurboAE models are trained at block length 64, rate 1/2, and $E_b/N_0 = 4$ dB, either through the analytic AWGN channel or through the condition-wise Sinkhorn AWGN surrogate. Curves are evaluated on the analytic AWGN channel over 30 seeds. The surrogate preserves the qualitative waterfall trend but remains above the analytic-channel training curve.